

and aligned in less than an hour. Full installation takes no more than half a day. So, FSO can be used in disaster management and emergency conditions. These systems employ powerful lasers that can transmit through glass and windows, thus further increasing installation options and equipment security. FSO technology solves last-mile bottlenecks in hours, not weeks—and without digging or infrastructure upgrades required by other PTP connections because it does not require expensive fibre-optic cable or regulatory spectrum licences for radio-frequency (RF) solutions. Thus, it offers a fast and high return on investment.

Free-space optical spectrum is nearly unlimited, thus very dense frequency re-use is possible that makes very convenient deployment of FSO systems worldwide. Also, FSO communication is immune to radiofrequency interference or saturation. FSO technology provides fibre-like availability with data rates up to 1.25Gbps (expandable up to 10Gbps) that does so with the low latency, low jitter, low bit error-rate connections.

This technology offers better interoperability with fibre networks and delivers scalable bandwidth supporting all protocols. Moreover, FSO technology's narrow beam transmission is typically two metres versus twenty metres and more for traditional, even newer radio-based technologies such as millimetre-wave radio. The lasers emit very sharp, focussed and narrow (small angle/small divergence) invisible light that results in very small spot size at a long distance (spot diameter (m) = angle (milliradians) × range (km)). This makes FSO hard to intercept, making it ideal for high-security applications such as financial, military and medical activities.

Fourth-generation FSO systems offer exclusive automatic power-level control, eliminating short-distance optical saturation found in first-generation systems. The latest fourth-generation FSO lasers are 100 per cent class 1M eye-safe technology. Also, they employ active tracking multiple-



Fig. 2: FSO working illustration

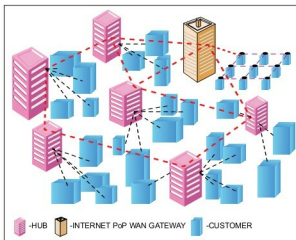


Fig. 3: The PTP, star, mesh and ring hybrid topologies for FSO

beam solution. Such systems eliminate mistargeting and provide parallel beam redundancy, solving problems such as a bird flying through the path of a first-generation FSO system.

FSO network topologies

Several network topologies are possible for FSO networks. PTP, star, ring and mesh are all feasible network architectures for laser communications. PTP topology is the simplest of the physical layouts of network devices. PTP connections mean that two devices (nodes) have a single path for data to travel between them with nothing breaking up that path.

On the other hand, in star connections, all devices are connected to a central hub utilising independent links. The central node is usually a hub or a multiplexer that utilises repeaters to forward data.

In a ring topology, all nodes are connected to one another in the shape of a closed loop, so that each node is

connected directly to two other nodes, one on either side of it.

In a mesh topology, devices are connected with many redundant interconnections between network nodes. Each node is connected with two to four nodes and is able to receive, transmit and forward data packets. All topologies comprise a combination of PTP links.

Additionally, hybrid topologies including ring, mesh, PTP and/or star interconnections have been proposed for FSO systems (Fig. 3). In ring interconnections, customer nodes can be ring nodes of more than one ring or one hop away from another ring. At the same time, PTP connections with nodes that do not belong to the ring can exist. On the other hand, in star interconnections, central nodes (hubs) are connected with other central nodes utilising PTP links. The primary reason of extending the standard network topologies is the increase of reliability and coverage area whilst keeping the associated cost low.

The choice of network topology significantly affects the performance, reliability, scalability, design complexity and overall cost of a free-space optics network.

Challenges

There are several signal propagation impediments for FSO that affect light propagation through free space. The turbulences in the signal may be due